Calc I Lecture 3: Introducing rates of change, and limits.

Let's now return to the problem we posed at the beginning:

If someone is driving a car from point A to point B, and we know their position at any point in time, how do we determine their speed?

We should distinguish between two kinds of speed:

- Average speed over some period of time $= \dfrac{\text{total distance}}{\text{time elapsed}}$
- Instantaneous speed at a specific time.

We are interested in the instantaneous speed.

Note: technically a more accurate term to use is velocity, rather than speed.

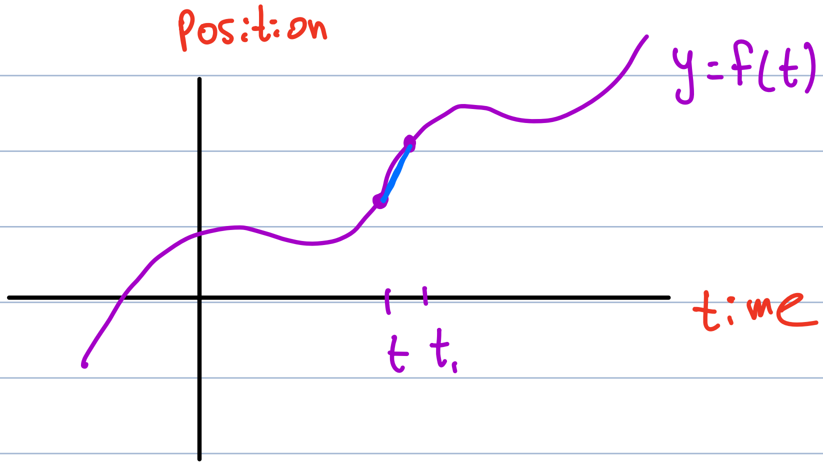
speed $= \dfrac{\text{"distance"}}{\text{time}}$

velocity: $\dfrac{\text{"displacement"}}{\text{time}}$

change in position, could be positive or negative

Idea: • To estimate instantaneous velocity, we zoom in very close to the time we're interested in and measure the average velocity over a short time interval.

- The true instantaneous velocity is the limiting value of these estimates as we zoom closer and closer.



Example: suppose position is given by

$$f(t) = t^3 + t - t^2, \qquad 0 \le t \le 2$$

Estimate instantaneous velocity at $t=1$

first approx: compare $t=1$ to $t=1.1$

$$\text{velocity} \approx \frac{\text{displacement}}{\text{change in time}} = \frac{f(1.1)-f(1)}{1.1-1} = 2.21$$

better approx: compare $t=1$ to $t=1.01$

$$\text{velocity} \approx \frac{f(1.01)-f(1)}{1.01-1} = 2.021$$

even better: compare $t=1$ to $t=1.001$

$$\text{velocity} \approx \frac{f(1.001)-f(1)}{1.001-1} = 2.002001$$

it looks like these estimates get closer and closer to 2

in fact, the "true" instantaneous velocity at $t=1$ is exactly 2. To explain formally, need to study...

Limits

Definition: Given a function $f(x)$ and a number a,

we write $$\lim_{x \to a} f(x) = L$$

or in words

"the limit of $f(x)$, as x approaches a, equals L"

if we can make the values of $f(x)$ arbitrarily close to L by taking x to be sufficiently close to a (on either side) but not equal to a.

Note: We can talk about $\lim_{x \to a} f(x)$ even if the function f is not defined at a. The limit tells us what the values of the function tend to as x approaches a.

Example: Consider the function $f(x) = \frac{\sin(x)}{x}$

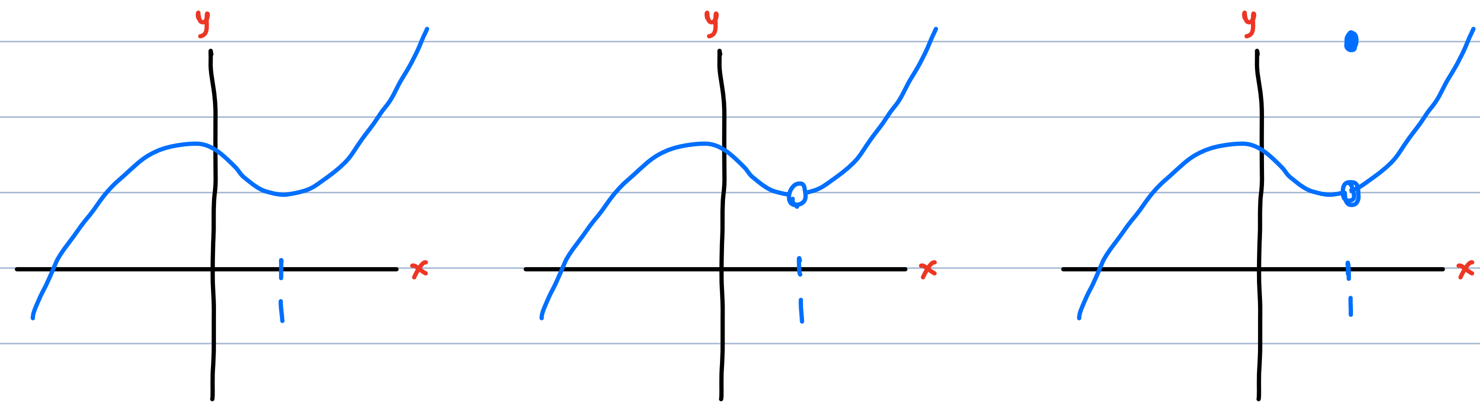
It is not defined at $x = 0$.

but it turns out that $\lim_{x \to 0} \frac{\sin(x)}{x} = 1$

as x approaches 0 from either side, $\frac{\sin(x)}{x}$ gets arbitrarily close to 1.

Note: the value of a limit $\lim_{x \to a} f(x)$ depends **only** on the behavior of the function **near** a. It doesn't matter what the value of the function is **at** a, or if it's even defined.

Example: the following graphs represent functions that all have the same limit as $x \to 1$



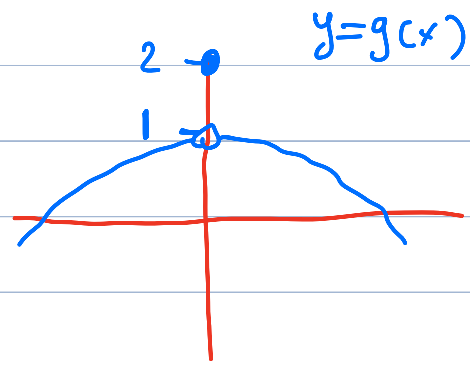
behave the same **near** 1 (just not at 1)

so $\lim_{x \to 1} f(x)$ is the same for all 3 functions

Example Suppose we know that $\lim_{x \to 0} \frac{\sin(x)}{x} = 1$

Now define another function $g(x)$ by

$$g(x) = \begin{cases} \frac{\sin(x)}{x} & \text{if } x \neq 0 \\ 2 & \text{if } x = 0 \end{cases}$$



What is $\lim_{x \to 0} g(x)$?

Answer: still 1

Note: In some cases $\lim_{x \to a} f(x)$ <u>does not exist</u>.

Here are a few examples:

Example: Suppose $f(x)$ is defined by $f(x) = \frac{x}{|x|}$

a) sketch the graph of f

b) Evaluate $\lim_{x \to 0} f(x)$ (write DNE if it does not exist).

c) What about $\lim_{x \to \frac{1}{2}} f(x)$?

a) Split into cases.

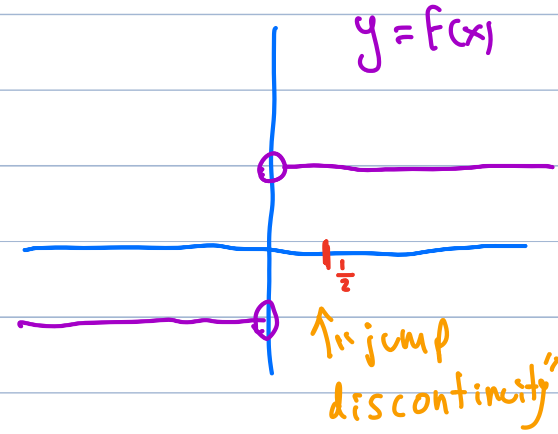
$x = 0$: $f(x)$ undefined (division by 0)

$x > 0$: $|x| = x \Rightarrow \frac{x}{|x|} = 1$

$x < 0$: $|x| = -x \Rightarrow \frac{x}{|x|} = -1$

$$f(x) = \begin{cases} 1 & \text{if } x > 0 \\ -1 & \text{if } x < 0 \end{cases}$$

undefined at 0



b) $\lim_{x \to 0} f(x)$ DNE

because f approaches 1 as x approaches 0 from right

f approaches -1 as x approaches 0 from left

$\Rightarrow$ inconsistent, no limit.

c) $\lim_{x \to \frac{1}{2}} f(x) = 1$ ($f = 1$ when x is near $\frac{1}{2}$ on either side) book calls it $[x]$

Example: The floor function $f(x) = \lfloor x \rfloor$ is defined by rounding down x to the nearest whole number below it (or equal)

e.g. $\lfloor \pi \rfloor = \lfloor 3.14159\ldots \rfloor = 3$

$\lfloor 2.99 \rfloor = 2$

$\lfloor -3.1 \rfloor = -4$

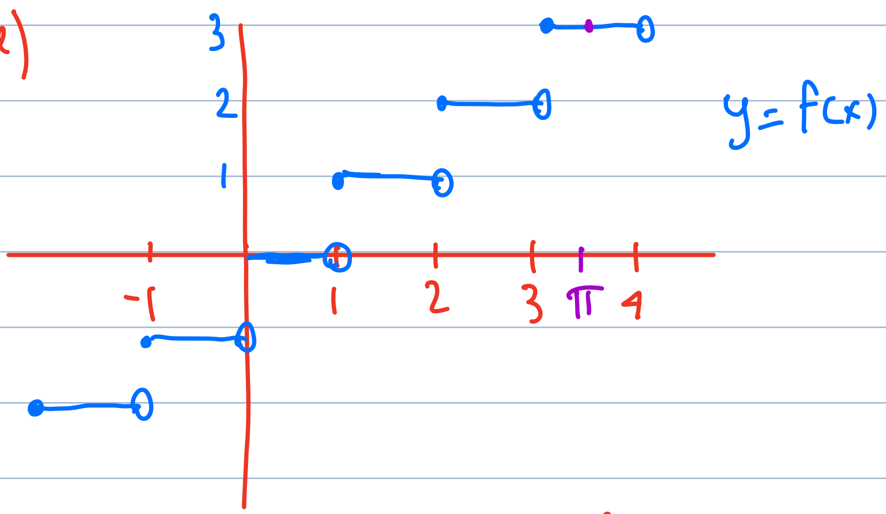
$\lfloor 3 \rfloor = 3$

a) sketch the graph of f

b) Evaluate the limit $\lim_{x \to 2} f(x)$ (write DNE if does not exist)

c) What about $\lim_{x \to \pi} f(x)$?

a)



b) $\lim_{x \to 2} f(x)$ DNE (same as before: approach 2 from right, 1 from left, inconsistent)

c) $\lim_{x \to \pi} f(x) = 3$ ($f(x) = 3$ near π on either side)

Example Consider the function $f(x) = \sin(\frac{1}{x})$

a) Sketch the graph of f

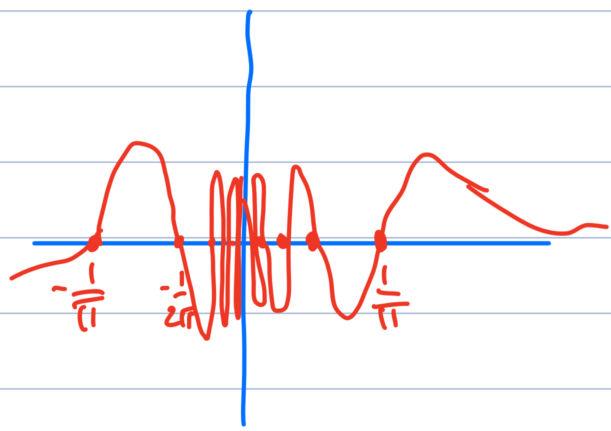
b) Evaluate the limit $\lim_{x \to 0} f(x)$

c) What about $\lim_{x \to \frac{1}{\pi}} f(x)$?

Question: when is $\sin(\frac{1}{x}) = 0$?

if $\frac{1}{x} = n\pi \Rightarrow x = \frac{1}{n\pi}$

$\{\underbrace{\frac{1}{\pi}, \frac{1}{2\pi}, \frac{1}{3\pi}, \dots}_{\text{get closer and closer to } 0}\}$



in between the graph oscillates
increasingly frequent oscillations of y between 1 and -1 as x gets close to 0

$\lim_{x \to 0} \sin(\frac{1}{x})$ $\boxed{\text{DNE}}$

because no matter how close x is to 0, the function continues to oscillate, does not approach a limiting value.

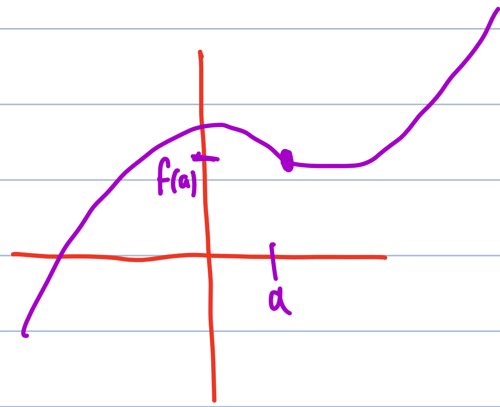
Example: let $f(x) = \frac{1}{x}$

as x approaches 0, $f(x)$ gets arbitrarily large

$\Rightarrow \lim_{x \to 0} f(x)$ DNE

<u>Observation</u>. • Examples where $\lim_{x\to a} f(x)$ <u>does not exist</u> exhibit "unusual" behavior: either there is a sudden jump at $x=a$ or the function oscillates around $x=a$ or function goes to infinity

• In contrast, if $f(x)$ is "nice" near $x=a$ (more precisely, if f is <u>continuous</u> at a, a concept we learn later) then $\lim_{x\to a} f(x) = f(a)$.



One-sided limits

Sometimes we want to analyze the behavior of a function on one side only.

<u>Definition</u>: Given a function $f(x)$ and a number a,

- we write

$$\lim_{x \to a^-} f(x) = L$$

or in words

"the limit of $f(x)$, as x approaches a <u>from the left</u>, equals L"

if we can make the values of $f(x)$ arbitrarily close to L by taking x to be sufficiently close to a and <u>less than</u> a.

- Analogously, we write

$$\lim_{x \to a^+} f(x) = L$$

or in words

"the limit of $f(x)$, as x approaches a <u>from the right</u>, equals L"

if we can make the values of $f(x)$ arbitrarily close to L by taking x to be sufficiently close to a and <u>greater than</u> a

Example: The graph of a function $f(x)$ is given below.

Given this graph, evaluate the limits (if they exist):

a) $\lim_{x \to 2^-} f(x)$ 3

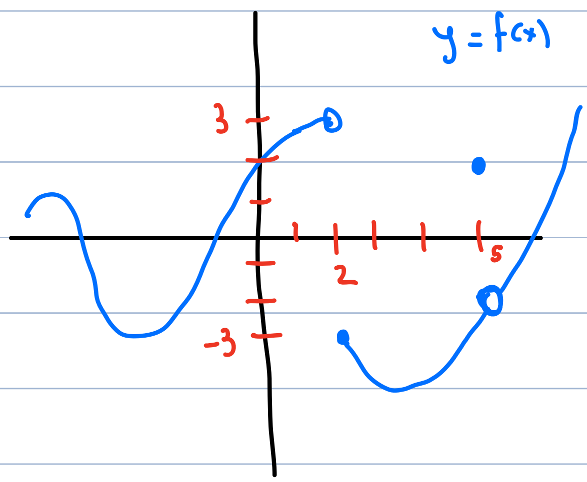
b) $\lim_{x \to 2^+} f(x)$ -3

c) $\lim_{x \to 2} f(x)$ DNE

d) $\lim_{x \to 5^-} f(x)$ -2

e) $\lim_{x \to 5^+} f(x)$ -2

f) $\lim_{x \to 5} f(x)$ -2



Observation: $\lim_{x \to a} f(x) = L$ if and only if

$\lim_{x \to a^-} f(x) =$ AND $\lim_{x \to a^+} f(x) =$

Sidenote: why are we studying such strange functions?

Infinite Limits

Definition: Given a function $f(x)$ and a number a

- we write

$$\lim_{x \to a} f(x) = \infty$$

or in words "$f(x)$ approaches infinity as x approaches a"
if the values of $f(x)$ can be made arbitrarily large by taking x sufficiently close to a (on either side) but not equal to a.

- Similarly, we write

$$\lim_{x \to a} f(x) = -\infty$$

or in words "$f(x)$ approaches negative infinity as x approaches a"
if the values of $f(x)$ can be made arbitrarily negative by taking x sufficiently close to a (on either side) but not equal

Caution: if $\lim_{x \to a} f(x) = \infty$ or $\lim_{x \to a} f(x) = -\infty$

we still say that "the limit of $f(x)$ as x approaches a" does not exist since infinity is not a number.
However, specifying an infinite limit gives more information about the behavior of the function.

<u>Example</u> In each case, evaluate whether $\lim_{x \to 0} f(x) = \infty$,

$\lim_{x \to 0} f(x) = -\infty$, or neither.

a) $f(x) = \dfrac{1}{x}$

b) $f(x) = \dfrac{1}{x^2}$

c) $f(x) = \dfrac{\sin\left(\frac{1}{x}\right)}{x}$

d) $f(x) = \dfrac{x-1}{x^2}$

<u>Note</u>: We can also define <u>one-sided infinite limits</u>:

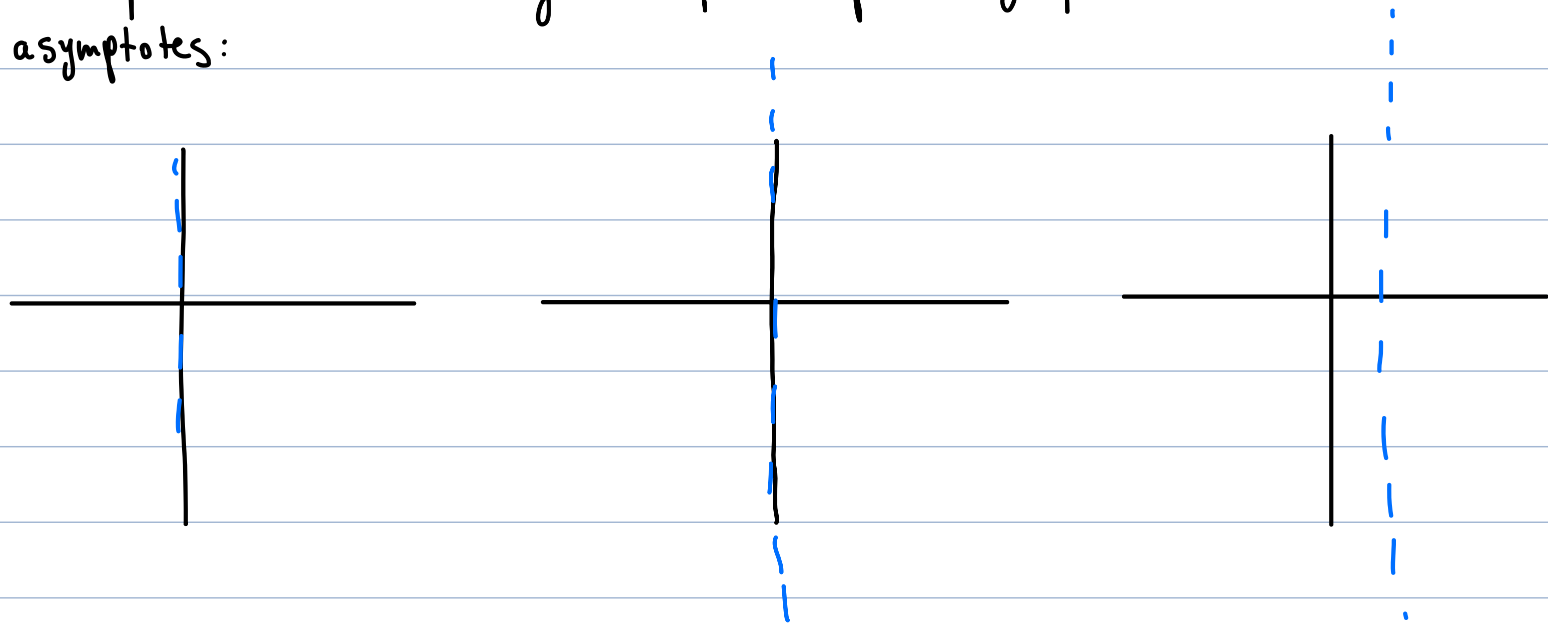
- $\lim_{x \to a^-} f(x) = \infty$ if as x approaches a from left, $f(x)$ goes to infinity) (arbitrarily large
- $\lim_{x \to a^-} f(x) = -\infty$ if as x approaches a from left, $f(x)$ goes to $-\infty$

$\lim_{x \to a^+} f(x) = \infty$ if . . .

$\lim_{x \to a^+} f(x) = -\infty$ if . . .

Definition: Given a function $f(x)$, we say that $x=a$ is a **vertical asymptote** of f if at least one of the one-sided limits is ∞ or $-\infty$.

Example: the following examples depict graphs with vertical asymptotes:



Note: vertical asymptotes generally arise at places where we divide by 0.

Example: Identify all the vertical asymptotes of $f(x) = \tan(x)$

Example: identify all vertical asymptotes of the function $f(x) = \dfrac{\ln(x)}{x-2}$

How to evaluate infinite limits?

Often the function will be of the form

$$f(x) = \frac{g(x)}{h(x)}$$

And potential asymptotes are points a where $h(a) = 0$.

Let's assume for now that $g(a) \neq 0$ (otherwise it's more complicated)

Then:

- Consider the sign of $g(a)$ and the sign of $h(x)$ when x is near and to the right of a to determine if $\lim_{x \to a^+} \frac{g(x)}{h(x)}$ is ∞ or $-\infty$
- Similarly for the case of x near and to the left of a to determine if $\lim_{x \to a^-} \frac{g(x)}{h(x)}$ is ∞ or $-\infty$
- If the two one-sided limits agree, then this is also $\lim_{x \to a} \frac{g(x)}{h(x)}$

Example: Suppose $f(x) = \frac{e^x}{(x-2)^3}$.

a) Find the infinite limit $\lim_{x \to 2^+} f(x)$

b) Find the infinite limit $\lim_{x \to 2^-} f(x)$

c) What can you say about $\lim_{x \to 2} f(x)$?

[illegible]